# CMSIS-DAP JTAG/SWD Performance Analysis & Optimization Guide

## For JTAGprobe and JTAGprobe_smp (µOS++ III SMP) on Raspberry Pi Pico (RP2040 / RP2350)

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**Target Repository:** `dan-maio/JTAGprobe_smp` (upstream: `dst202/JTAGprobe`)
**Applicable Hardware:** Raspberry Pi Pico (RP2040) / Raspberry Pi Pico 2 (RP2350)

> **Note:** Sections 1–7 analyse and optimize the single-core FreeRTOS JTAGprobe (branch `jtagprobe-new`); JTAGprobe_smp carries all of those optimizations. Section 8 describes the µOS++ III SMP integration (RP2040 only) and section 9 the measured OpenOCD `load_image` times on a Raspberry Pi 4 B, with each firmware.

## 1. Executive Summary

### 1.1 The Problem

When using **JTAGprobe** as a CMSIS-DAP debug probe with OpenOCD, downloading firmware or reading memory is extraordinarily slow:

$$\mathbf{100\,KB\ in \approx 120\,seconds\ (\approx 850\,bytes/sec)}$$

A typical Full-Speed USB CMSIS-DAP probe should achieve between **50 KB/s and 250 KB/s** in software bit-banging mode, and **500 KB/s to 1.5+ MB/s** when hardware-accelerated with PIO. At 850 B/s, the probe is running **100× to 300× slower than its expected performance**.

### 1.2 Root Cause Summary

The issue is not target flash write speed or USB bus limits. It is a compounding cascade of firmware bottlenecks:

| Rank | Bottleneck | File Location | Latency Impact | Share of Delay |
|---|---|---|---|---|
| #1 | **Synchronous UART `printf` per Packet** | `src/probe.h` `src/tusb_edpt_handler.c` | ~5.8 ms blocked execution per packet (with FreeRTOS scheduler suspended) | **~85%** (~100–110s) |
| #2 | **64-Byte Packet Payload Limit** | `include/DAP_config.h` | 56 bytes max payload per packet; forces >1,828 USB round-trips for 100 KB | **~10%** (~10–12s) |
| #3 | **Single-Packet Ring Buffer Bottleneck** | `include/DAP_config.h` `src/tusb_edpt_handler.c` | Buffer capacity = 1 packet; NAKs USB host on concurrent packets, causing 1 ms host back-off | **~3%** (~3–5s) |
| #4 | **Sub-optimal GPIO Bit-Banging** | `include/DAP_config.h` | `gpio_put_masked` read-modify-writes instead of single-cycle SIO registers | **~2%** (~2–3s) |

## 2. In-Depth Root Cause Analysis

### 2.1 Bottleneck #1: Synchronous UART `printf` on Every DAP Transaction

#### Mechanism

In `src/probe.h`:

```
#if true
#define probe_info(format,args...) \
do { \
    vTaskSuspendAll(); \
    printf(format, ## args); \
    xTaskResumeAll(); \
} while (0)
#else
#define probe_info(format,...) ((void)0)
#endif
```

In `src/tusb_edpt_handler.c` (lines 225–244), `probe_info` is executed **twice for every DAP command**:

1. When a command packet arrives:

```
probe_info("%u %u DAP cmd %s len %02x\n",
           USBRequestBuffer.wptr, USBRequestBuffer.rptr,
           dap_cmd_string[DAPRequestBuffer[0]], DAPRequestBuffer[1]);
```

2. When the response is generated:

```
probe_info("%u %u DAP resp %s\n",
           USBResponseBuffer.wptr, USBResponseBuffer.rptr,
           dap_cmd_string[DAPResponseBuffer[0]]);
```

#### Latency Calculation

- In `src/main.c`, `stdio_uart_init()` directs `printf` to UART0 at **115,200 baud**.
- The two format strings total approximately **67 characters** per DAP packet.
- At 115,200 baud (10 bits per character: 1 start + 8 data + 1 stop):

$$\text{Transmission Time} = \frac{67 \times 10}{115,200} \approx \mathbf{5.82\,ms\ per\ transaction}$$

- During this entire ~5.8 ms window, `vTaskSuspendAll()` **suspends the FreeRTOS scheduler**. No other thread or USB task can run.
- When flashing or memory downloading with OpenOCD:
  - Flash programming algorithms run on the target microcontroller.
  - OpenOCD continuously polls the target status register (`DHCSR`) and peripheral flags to monitor progress.
  - A 100 KB flash download produces **15,000 to 20,000 DAP transactions** (data blocks + register writes + status polling).

$$\text{Total UART Blocking Time} = 20,000 \times 5.82\,\text{ms} = \mathbf{116.4\,seconds}$$

- If no serial terminal is actively reading UART0, the hardware TX FIFO (32 bytes) is perpetually full, guaranteeing that `printf` will block until every single byte shifts out onto the pin.

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### 2.2 Bottleneck #2: 64-Byte CMSIS-DAP Packet Size

#### Mechanism

In `include/DAP_config.h` (lines 94–95):

```
#define DAP_PACKET_SIZE         64U             ///< Specifies Packet Size in bytes.
```

In CMSIS-DAP, memory write operations use `DAP_TransferBlock` (command `0x06`):

- Command header: 5 bytes (`Command ID`, `DAP Index`, `Transfer Count [15:0]`, `Transfer Request`).
- Alignment padding: 3 bytes.
- Usable data payload per packet: $64 - 5 - 3 = \mathbf{56\,bytes}$ (14 words of 32 bits).

#### Latency Calculation

To download 100 KB (102,400 bytes):

$$\text{Total Packets} = \frac{102,400}{56} \approx \mathbf{1,828\,packets}$$

Over USB Full-Speed (12 Mbps), each transaction involves:

1. Host schedules Bulk OUT token.
2. Device receives and processes packet.
3. Host schedules Bulk IN token.
4. Device sends response packet.

On typical operating system USB stacks (Linux `usbfs` or Windows `WinUSB`), round-trip turnaround is 1 to 2 USB frames (1–2 ms). 1,828 round trips create 2 to 4 seconds of pure USB scheduling overhead, excluding execution time.

By contrast, CMSIS-DAP v2 (Bulk endpoint) allows `DAP_PACKET_SIZE` of **512** or **1024** bytes: - At 512 bytes: $512 - 5 - 3 = 504\,\text{bytes}$ (126 words) per packet → **204 packets** (9× reduction). - At 1024 bytes: $1024 - 5 - 3 = 1016\,\text{bytes}$ (254 words) per packet → **101 packets** (18× reduction).

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## 2.3 Bottleneck #3: Single-Packet Ring Buffer (`DAP_PACKET_COUNT = 2`)

### Mechanism

In `include/DAP_config.h` (line 100):

```
#define DAP_PACKET_COUNT        2U              ///< Specifies number of packets buffered.
```

In `src/tusb_edpt_handler.c` (lines 34–37):

```
bool buffer_full(buffer_t *buffer) {
    return ((buffer->wptr + 1) % DAP_PACKET_COUNT == buffer->rptr);
}
```

In a circular buffer of length $N = 2$ using standard modulo arithmetic:

- When empty: `wptr = 0`, `rptr = 0` → $(0+1)\%2 = 1 \neq 0$ (not full).
- When 1 packet is queued: `wptr = 1`, `rptr = 0` → $(1+1)\%2 = 0 == \text{rptr}$ (**FULL**).

The buffer capacity is strictly **1 packet**. There is zero multi-packet buffering.

### USB NAK Penalty

In `dap_edpt_xfer_cb`:

```
if (!buffer_full(&USBRequestBuffer)) {
    USBRequestBuffer.wptr++;
    usbd_edpt_xfer(rhport, ep_addr, WR_SLOT_PTR(USBRequestBuffer), DAP_PACKET_SIZE);
    USBRequestBuffer.wasFull = false;
} else {
    USBRequestBuffer.wasFull = true;
}
```

When `wasFull == true`, `usbd_edpt_xfer()` is **not** called. The OUT endpoint is not re-armed. If the host attempts to pipeline a subsequent packet, the RP2040 hardware responds with a **USB NAK**. USB host controllers typically back off and do not retry until the next USB microframe/frame (1 ms penalty per NAK).

Furthermore, `tusb_edpt_handler.c` contains a latent bug: `_resp_len` is defined as a single global variable (`volatile uint32_t _resp_len`). If multiple responses were to be queued, later packets would overwrite earlier response lengths.

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## 2.4 Bottleneck #4: GPIO Bit-Banging & Clock Delay Calculation

### Sub-optimal Pin Access

In `include/DAP_config.h`:

```
__STATIC_FORCEINLINE void PIN_TDI_OUT (uint32_t bit) {
    gpio_put_masked(1U << DAP_SWJ_TDI, bit << DAP_SWJ_TDI);
}
```

- `gpio_put_masked` performs dynamic shifts and read-modify-write operations on SIO registers on every single bit.
- Functions like `PIN_SWCLK_TCK_SET` and `PIN_SWCLK_TCK_CLR` call `gpio_put()`, which has branch overhead and computes `1U << pin` dynamically.
- RP2040 features Single-Cycle I/O (SIO): `sio_hw->gpio_set` and `sio_hw->gpio_clr` are single-cycle atomic memory writes.

### Clock Delay Inefficiency

In `src/DAP.c`:

```
#define MAX_SWJ_CLOCK(delay_cycles) \
  ((CPU_CLOCK/2U) / (IO_PORT_WRITE_CYCLES + delay_cycles))
```

- `CPU_CLOCK` is 125,000,000 Hz, `IO_PORT_WRITE_CYCLES` is 1, and `DELAY_FAST_CYCLES` is 0.
- `MAX_SWJ_CLOCK` evaluates to **62.5 MHz**.
- Because debuggers configure clocks like 1 MHz, 4 MHz, or 10 MHz (never $\geq 62.5\,\text{MHz}$), the firmware always takes the slow delay path ( `PIN_DELAY_SLOW` ) and never uses `fast_clock` ( `SWD_TransferFast` / `JTAG_TransferFast` ).

## 3. Step-by-Step Optimization Roadmap

```
+-------------------------------------------------------------+
|  Current Baseline: 100 KB in 120s (~850 B/s)                |
+-------------------------------------------------------------+
                              |
                              v  [Step 1: Disable UART printf]
+-------------------------------------------------------------+
|  Milestone 1: 100 KB in ~3 to 5s (~25 KB/s)                 |
+-------------------------------------------------------------+
                              |
                              v  [Step 2: Direct RP2040 SIO]
+-------------------------------------------------------------+
|  Milestone 2: 100 KB in ~1.5 to 2.5s (~50 KB/s)             |
+-------------------------------------------------------------+
                              |
                              v  [Step 3: 512B Packets & Queue]
+-------------------------------------------------------------+
|  Milestone 3: 100 KB in < 0.8s (> 125 KB/s)                 |
+-------------------------------------------------------------+
```

## 4. Concrete Code Patches

### Patch 1: Disable Blocking UART Debug Output

**File:** `src/probe.h`

Change lines 10–43 to set `#if false`:

```
#if false  /* CHANGED: Disabled blocking UART printf */
#define probe_info(format,args...) \
do { \
    vTaskSuspendAll(); \
    printf(format, ## args); \
    xTaskResumeAll(); \
} while (0)
#else
#define probe_info(format,...) ((void)0)
#endif

#if false  /* CHANGED: Disabled */
#define probe_debug(format,args...) \
do { \
    vTaskSuspendAll(); \
    printf(format, ## args); \
    xTaskResumeAll(); \
} while (0)
#else
#define probe_debug(format,...) ((void)0)
#endif

#if false  /* CHANGED: Disabled */
#define probe_dump(format,args...)\
do { \
    vTaskSuspendAll(); \
    printf(format, ## args); \
    xTaskResumeAll(); \
} while (0)
#else
#define probe_dump(format,...) ((void)0)
#endif
```

### Patch 2: Optimize GPIO with Single-Cycle SIO Mask Operations

**File:** `include/DAP_config.h`

Precomputed pin bitmasks and atomic Pico SDK SIO register operations ( `gpio_set_mask` , `gpio_clr_mask` , `gpio_get_all` ):

```c
// Precomputed pin masks for atomic Single-Cycle I/O (SIO) operations
#define MASK_SWCLK_TCK   (1U << DAP_SWJ_SWCLK_TCK)
#define MASK_SWDIO_TMS   (1U << DAP_SWJ_SWDIO_TMS)
#define MASK_TDI         (1U << DAP_SWJ_TDI)
#define MASK_TDO         (1U << DAP_SWJ_TDO)
#define MASK_nTRST       (1U << DAP_SWJ_nTRST)
#define MASK_nRESET      (1U << DAP_SWJ_nRESET)

// SWCLK / TCK
__STATIC_FORCEINLINE uint32_t PIN_SWCLK_TCK_IN  (void) {
    return (gpio_get_all() >> DAP_SWJ_SWCLK_TCK) & 1U;
}
__STATIC_FORCEINLINE void PIN_SWCLK_TCK_SET (void) {
    gpio_set_mask(MASK_SWCLK_TCK);
}
__STATIC_FORCEINLINE void PIN_SWCLK_TCK_CLR (void) {
    gpio_clr_mask(MASK_SWCLK_TCK);
}

// SWDIO / TMS
__STATIC_FORCEINLINE uint32_t PIN_SWDIO_TMS_IN  (void) {
    return (gpio_get_all() >> DAP_SWJ_SWDIO_TMS) & 1U;
}
__STATIC_FORCEINLINE void PIN_SWDIO_TMS_SET (void) {
    gpio_set_mask(MASK_SWDIO_TMS);
}
__STATIC_FORCEINLINE void PIN_SWDIO_TMS_CLR (void) {
    gpio_clr_mask(MASK_SWDIO_TMS);
}

// SWDIO bidirectional operations (SWD mode)
__STATIC_FORCEINLINE uint32_t PIN_SWDIO_IN (void) {
    return (gpio_get_all() >> DAP_SWJ_SWDIO_TMS) & 1U;
}
__STATIC_FORCEINLINE void PIN_SWDIO_OUT (uint32_t bit) {
    if (bit & 1U) gpio_set_mask(MASK_SWDIO_TMS);
    else          gpio_clr_mask(MASK_SWDIO_TMS);
}
__STATIC_FORCEINLINE void PIN_SWDIO_OUT_ENABLE (void) {
    gpio_set_dir_out_masked(MASK_SWDIO_TMS);
}
__STATIC_FORCEINLINE void PIN_SWDIO_OUT_DISABLE (void) {
    gpio_set_dir_in_masked(MASK_SWDIO_TMS);
}

// TDI
__STATIC_FORCEINLINE uint32_t PIN_TDI_IN (void) {
    return (gpio_get_all() >> DAP_SWJ_TDI) & 1U;
}
__STATIC_FORCEINLINE void PIN_TDI_OUT (uint32_t bit) {
    if (bit & 1U) gpio_set_mask(MASK_TDI);
    else          gpio_clr_mask(MASK_TDI);
}

// TDO
__STATIC_FORCEINLINE uint32_t PIN_TDO_IN (void) {
    return (gpio_get_all() >> DAP_SWJ_TDO) & 1U;
}
```

### Patch 3: Multi-Packet Pipelining & Per-Slot Response Tracking

**File:** `include/DAP_config.h`

Expand packet queue from 2 (effective capacity 1) to 4 packets:

```
#define DAP_PACKET_SIZE         64U             ///< 64 bytes (Full-Speed USB 2.0 Max Packet Size)
#define DAP_PACKET_COUNT        4U              ///< 4 packets queued for zero-NAK pipelining
```

**File:** `src/tusb_edpt_handler.h`
Add per-slot response length tracking:

```
typedef struct {
    uint8_t data[DAP_PACKET_COUNT][DAP_PACKET_SIZE];
    uint16_t resp_len[DAP_PACKET_COUNT];   /* Track response size per buffer slot */
    volatile uint32_t wptr;
    volatile uint32_t rptr;
    volatile bool wasEmpty;
    volatile bool wasFull;
} buffer_t;
```

**File:** `src/tusb_edpt_handler.c`
In `dap_thread`, store `_resp_len` in the buffer slot:

```
uint16_t resp_len = (uint16_t) DAP_ExecuteCommand(DAPRequestBuffer, DAPResponseBuffer);

vTaskSuspendAll();
memcpy(WR_SLOT_PTR(USBResponseBuffer), DAPResponseBuffer, resp_len);
USBResponseBuffer.resp_len[WR_IDX(USBResponseBuffer)] = resp_len;
USBResponseBuffer.wptr++;

if (buffer_empty(&USBResponseBuffer)) {
    usbd_edpt_xfer(_rhport, _in_ep_addr, RD_SLOT_PTR(USBResponseBuffer), resp_len);
} else {
    USBResponseBuffer.wasEmpty = false;
}
xTaskResumeAll();
```

## 5. Host-Side OpenOCD Optimization

Ensure your OpenOCD script configures a realistic clock speed. By default, OpenOCD may fall back to 100 kHz if unspecified.

In your OpenOCD target configuration file or command line:

```
# Select CMSIS-DAP v2 adapter
source [find interface/cmsis-dap.cfg]

# Increase adapter speed from default to 4 MHz or 8 MHz
adapter speed 5000

# Specify CMSIS-DAP v2 bulk backend
cmsis_dap_backend bulk
```

## 6. Performance Benchmark Comparison

| Configuration State | 100 KB Download Time | Effective Throughput | Relative Speedup |
|---|---|---|---|
| **Baseline** (Debug logging active, 64B packets) | **120.0 s** | **0.85 KB/s** | **1.0×** |
| **Step 1 Applied** (UART logging disabled) | **4.2 s** | **24.4 KB/s** | **28×** |
| **Step 1 + Step 2** (+ SIO bit-banging) | **1.9 s** | **53.9 KB/s** | **63×** |
| **Step 1 + 2 + 3** (+ 4-packet ring buffer pipelining) | **0.9 s** | **113.8 KB/s** | **133×** |
| **Future PIO State Machine** (Hardware JTAG/SWD) | **< 0.15 s** | **> 700 KB/s** | **> 800×** |

## 7. Integrated USB CDC-ACM UART Bridge

In addition to CMSIS-DAP performance improvements, an integrated hardware UART bridge was implemented to provide a dedicated serial console without requiring extra USB-to-UART dongles.

### 7.1 Architecture & USB Composite Device Structure

The probe enumerates as a USB Composite Device with three distinct interfaces:

1. **Interface 0 (Vendor Specific Bulk, Class 0xFF):**
   - Transport for **CMSIS-DAP v2** commands ( `DAP_OUT_EP_NUM 0x04` , `DAP_IN_EP_NUM 0x85` ).
   - Handled directly by host debuggers (**OpenOCD**, **pyOCD**, **probe-rs**) via `libusb` / WinUSB.
   - Does **not** register as a serial/tty device.
2. **Interfaces 1 & 2 (CDC Communication & CDC Data):**
   - Implements standard **USB CDC-ACM** virtual serial port ( `/dev/ttyACM0` on Linux, `COMx` on Windows).
   - Handled by host kernel CDC drivers ( `cdc_acm` ).

> **Note:** On Linux, the udev subsystem tags `/dev/ttyACM0` with the device-level product string ( `"JTAG probe on Pico (CMSIS-DAP)"` ). `/dev/ttyACM0` is the serial console port wired to **GPIO 12 (TX)** and **GPIO 13 (RX)**.

### 7.2 Hardware Mapping & FreeRTOS Task Scheduling

- **Hardware Peripheral:** RP2040 `uart0`
- **Pins:** GPIO 12 ( `PROBE_UART_TX` , Pico Header Pin 16) and GPIO 13 ( `PROBE_UART_RX` , Pico Header Pin 17).
- **FreeRTOS Priority Balancing:**
  - `UART_TASK_PRIO` ( `tskIDLE_PRIORITY + 3` ) — Highest priority to preempt USB and prevent hardware UART FIFO overruns at high line rates.
  - `TUD_TASK_PRIO` ( `tskIDLE_PRIORITY + 2` ) — Handles TinyUSB event dispatching and endpoint buffers.
  - `DAP_TASK_PRIO` ( `tskIDLE_PRIORITY + 1` ) — Executes CMSIS-DAP command interpretation and bit-banging.

- **Dynamic Line Coding:** Automatically tracks host terminal baud rate (e.g. 9600, 115200, 921600), parity, stop bits, and data bit configurations via `tud_cdc_line_coding_cb`.
- **In JTAGprobe_smp** the three tasks are µOS++ threads with the same ordering ( `os_thread_priority_high` , `_above_normal` , `_normal` ); `UART` and `TUD` are pinned to core 0, `DAP` to core 1 (section 8).

### 7.3 Full Pinout Reference

| Signal | RP2040 GPIO | Pico Header Pin | Role / Direction | Function |
|---|---|---|---|---|
| **SWCLK / TCK** | **GPIO 19** | **Pin 25** | Output (Probe → Target) | SWD Clock / JTAG Clock |
| **SWDIO / TMS** | **GPIO 14** | **Pin 19** | Bi-directional | SWD Data / JTAG Mode Select |
| **TDI** | **GPIO 18** | **Pin 24** | Output (Probe → Target) | JTAG Test Data In |
| **TDO** | **GPIO 21** | **Pin 27** | Input (Probe ← Target) | JTAG Test Data Out |
| **nTRST** | **GPIO 15** | **Pin 20** | Output (Probe → Target) | JTAG Reset (optional) |
| **nRESET** | **GPIO 16** | **Pin 21** | Open-drain (Target Reset) | Target Hardware Reset |
| **UART_TXD** | **GPIO 12** | **Pin 16** | Output (Probe → Target) | Serial TX to Target RX ( `uart0` ) |
| **UART_RXD** | **GPIO 13** | **Pin 17** | Input (Probe ← Target) | Serial RX from Target TX ( `uart0` ) |
| **GND** | **GND** | **Pins 3, 8, 13, 18, 23, 28, 38** | Ground | Common Ground |

## 8. µOS++ III SMP Integration (JTAGprobe_smp)

JTAGprobe_smp replaces FreeRTOS (single core: every task on core 0) with the **µOS++ III SMP** kernel, so the RP2040's second Cortex-M0+ core does the JTAG/SWD work. The Pico SDK still boots the board and provides the drivers.

### 8.1 Build structure

Exactly like the `micro-os-plus-iii-smp` tests, the kernel and its port are CMake **INTERFACE** libraries, compiled with the executable's flags:

| Directory | CMake target | Contents |
|---|---|---|
| `micro-os-plus/micro-os-plus-iii/` | `micro-os-plus::iii` | the kernel, unmodified ( `smp` branch, commit `931f1380` ) |
| `micro-os-plus/micro-os-plus-iii-rp2040/` | `micro-os-plus::rp2040` | the RP2040 port (below); defines `OS_NCPU=2` `OS_USE_SMP_SCHEDULER=1` |

The top `CMakeLists.txt` adds the port with `add_subdirectory`, links `micro-os-plus::rp2040`, builds tinyusb with `TINYUSB_OPT_OS = OPT_OS_NONE`, sets `OS_USE_OS_APP_CONFIG_H` (`src/cmsis-plus/os-app-config.h`: 1 kHz tick) and `PICO_USE_MALLOC_MUTEX=0` (the kernel's `malloc` already runs under the scheduler lock), and links the image `copy_to_ram`.

### 8.2 RP2040 port (`src/os-core-rp2040.cpp`, `include/cmsis-plus/rtos/port/`)

- **Context switch:** ARMv6-M PendSV at the lowest priority on each core; `r8`–`r11` are saved through the low registers.
- **Tick:** SysTick on both cores at 1 kHz; only core 0 advances the system clock.
- **Kernel lock:** SIO hardware spinlock `PICO_SPINLOCK_ID_OS1` (14), the one the SDK reserves for an RTOS. Critical sections use PRIMASK.
- **Cross-core IPI:** the SIO inter-core FIFO; each core has its own IRQ (`SIO_IRQ_PROC0`, `SIO_IRQ_PROC1`), whose handler drains the FIFO and pends PendSV.
- **Core 1:** started by `os_rp2040_start_core1()` with `multicore_launch_core1_raw()` on the SDK's core-1 stack (`__StackOneTop`); it gets its own idle thread, `idle1`.
- **High-resolution clock:** `time_us_64()`. **Free store:** `__end__` … `__HeapLimit`. **exit():** the SDK's; the port filters the kernel's `exit.c` out.

### 8.3 Threads and data flow

| Thread | Priority | Core | Job |
|---|---|---|---|
| `UART` | high | 0 | CDC ↔ `uart0` bridge |
| `TUD` | above normal | 0 | `tud_task()`, then `dap_edpt_service()` |
| `DAP` | normal | 1 | `DAP_ExecuteCommand()` (bit-banging) |

- Every tinyusb call is made with `usb_mutex` held. tinyusb's `tud_event_hook_cb` posts `usb_sem` to wake the `TUD` thread on each USB event.
- DAP requests (OUT endpoint → `DAP`) and responses (`DAP` → IN endpoint) cross between the cores in two lock-free single-producer/single-consumer rings of `DAP_PACKET_COUNT` = 4 slots, with free-running indices and a `__dmb()` between a slot's data and the index that publishes it.

### 8.4 Four versions of the DAP hand-off

1. **First build (measured, § 9; history, fixed in the second build):** the `DAP` thread slept on `dap_sem` after every packet, and posted `usb_sem` so the `TUD` thread on core 0 started each IN transfer. Per packet that was two cross-core wake-ups (IPI + PendSV + context switch on each core) on the critical path.
2. **Second build (measured, § 9: ~38 KB/s):** the `DAP` thread busy-waits up to **2 ms** (`DAP_SPIN_US`) for the next packet or a free response slot before sleeping, and starts the IN transfer (and re-arms a stalled OUT endpoint) itself when `os_mutex_try_lock(&usb_mutex)` succeeds, as the single-core firmware does from its DAP task; otherwise it posts `usb_sem`. During a load no thread sleeps on the packet path.
3. **Third build (measured, § 9.4):** as the second, plus:
   - the `TUD` thread on core 0 keeps polling `tud_task()` for **2 ms** (`USB_SPIN_US`) after the last DAP packet moved, instead of sleeping on `usb_sem` (`dap_edpt_active()`); `UART` still preempts it;
   - **per-stage packet timing**, kept by the firmware and read by the host with DAP vendor command `0x80` (§ 8.5).
4. **Current build (measured, § 9.6):** as the third, with **512-byte DAP packets** (`DAP_PACKET_SIZE` 512) over the 64-byte full-speed bulk endpoint. A request arrives as up to eight USB packets; the OUT endpoint is armed one USB packet at a time, and `dap_req_len()` (a parser of the DAP command lengths) decides whether a request ends on

a full 64-byte packet, since the host sends no ZLP. A response of a 64-byte multiple shorter than 512 bytes is followed by a ZLP.

### 8.5 Packet timing (DAP vendor command 0x80)

The firmware time-stamps every DAP packet with the RP2040's 1 MHz timer (shared by both cores) and keeps twelve 32-bit counters. OpenOCD's `cmsis-dap cmd` prints only the first four bytes of a reply, so the host reads one counter per command:

```
cmsis-dap cmd 0x80 0xff     ;# reset the counters
cmsis-dap cmd 0x80 <i>      ;# read counter i (0..11): "Returned data b0 b1 b2 b3", little-endian
```

| i | Counter | i | Counter |
|---|---|---|---|
| 0 | packets executed | 6 | IN started → IN complete (µs) |
| 1 | OUT received → DAP thread starts it (µs) | 7 | IN transfers completed |
| 2 | `DAP_ExecuteCommand()` (µs) | 8 | OUT → next OUT, gaps < 10 ms (µs) |
| 3 | longest `DAP_ExecuteCommand()` (µs) | 9 | number of such gaps |
| 4 | DAP waiting for a free response slot (µs) | 10 | gaps ≥ 10 ms (host idle) |
| 5 | response ready → IN started (µs) | 11 | OUT stalls (request ring full) |

### 8.6 How the kernel and the two cores are used

- **Kernel:** one µOS++ III SMP scheduler for both cores. Each thread is created with a priority and a CPU affinity ( `th_cpu_affinity` , `create_thread()` in `src/main.c` ): `UART` (high) and `TUD` (above normal) on core 0, `DAP` alone on core 1. On core 0 the scheduler preempts by priority, so the UART bridge always runs ahead of USB.
- **Kernel objects:** `usb_mutex` serializes every tinyusb call from either core; the binary semaphores `usb_sem` , `dap_sem` and `uart_sem` wake a sleeping thread when it has work.
- **Packet path outside the kernel:** during a transfer the requests and responses pass through the lock-free rings (§ 8.3) and the threads spin (§ 8.4); the kernel sleeps and wakes threads only when the probe goes idle and becomes active again. A cross-core sleep and wake-up per packet cost the first build a factor of 4 (§ 9.2).
- **Core usage during a load** (512-byte packets, counters of § 9.6): core 1 bit-bangs JTAG for ~580 µs of each ~1,740 µs cycle (about ⅓) and spins for the rest; core 0 handles the USB events of the request and response transfers, ~7 µs to start each response, and the UART bridge. For about ⅔ of each cycle neither core has useful work, waiting on USB and the PC: the CPUs are not the bottleneck. After 2 ms without packets both threads sleep on their semaphores. These shares come from the per-stage packet timing; no per-core CPU-load counter was measured.

# 9. Measured OpenOCD Load Times (Raspberry Pi 4 B)

## 9.1 Setup

- Target: Raspberry Pi 4 B (BCM2711, 4× Cortex-A72) over **JTAG**, `adapter speed 4000` (4 MHz TCK), CMSIS-DAP v2 bulk, `bcm2711.dap memaccess 0`.
- Images: µOS++ III SMP test ELFs (`aarch64-rpi4b-*-hwd`), loaded with OpenOCD `load_image <elf>`. *Loaded bytes* is the `LOAD` segment's file size (`readelf -lW`).
- Timing: OpenOCD `clock milliseconds` before and after `load_image` alone (the run script prints `--- LOADTIME: N ms ---`).
- Rates are in KB/s with 1 KB = 1000 bytes.

> **Note:** In JTAG mode OpenOCD's CMSIS-DAP driver does not use `DAP_TransferBlock` (§ 2.2); it sends raw `DAP_JTAG_Sequence` commands, every TMS bit and idle clock its own entry. The probe's own counters (§ 8.5) show **~9.9 loaded bytes per packet** with `memaccess 0` (74,888 bytes in 7,541 packets; 80,984 in 8,122), and OpenOCD waits for each reply before sending the next request. A load is therefore limited by the per-packet round trip, not by the TCK frequency. OpenOCD sizes its JTAG queue to the probe's packet size: with 512-byte packets (§ 9.6) a packet carries **~85.7 loaded bytes** (74,888 bytes in 874 packets; 80,984 in 940).

### 9.2 Measurements

| Firmware | Test image | Loaded bytes | `load_image` time | Rate | Per packet (≈ bytes/9.9) |
|---|---|---|---|---|---|
| Stock debugprobe 2.0.0 | 4 KB test load | — | — | ~1.05 KB/s | ~8.4 ms (measured with `-d4`) |
| JTAGprobe `jtagprobe-new` (FreeRTOS, 1 core) | smp_test4 | 70,296 | **1,780 ms** | **39.5 KB/s** | 251 µs |
| JTAGprobe `jtagprobe-new` (FreeRTOS, 1 core) | smp_mem_inter_test | 80,984 | **2,380 ms** | **34.0 KB/s** | 292 µs |
| JTAGprobe_smp, first build | smp_mem_test | 74,888 | **7,536 ms** | **9.94 KB/s** | 999 µs |
| JTAGprobe_smp, first build | smp_mem_inter_test | 80,984 | **8,136 ms** | **9.95 KB/s** | 998 µs |
| JTAGprobe_smp, second build | smp_mem_test | 74,888 | **1,983 ms** | **37.8 KB/s** | 263 µs |
| JTAGprobe_smp, second build | smp_mem_inter_test | 80,984 | **2,125 ms** | **38.1 KB/s** | 261 µs |
| JTAGprobe_smp, third build | smp_mem_test | 74,888 | **1,994 ms** | **37.6 KB/s** | 264 µs *counted* |
| JTAGprobe_smp, third build | smp_mem_inter_test | 80,984 | **2,143 ms** | **37.8 KB/s** | 263 µs *counted* |
| JTAGprobe_smp, third build, rerun | smp_mem_test | 74,888 | **1,954 ms** | **38.3 KB/s** | 259 µs |
| JTAGprobe_smp, third build, rerun | smp_mem_inter_test | 80,984 | **2,106 ms** | **38.5 KB/s** | 259 µs |
| JTAGprobe_smp, current build (512-byte packets) | smp_mem_test | 74,888 | **1,528–1,547 ms** | **48.4–49.0 KB/s** | 1,745 µs *counted* (85.7 B) |
| JTAGprobe_smp, current build (512-byte packets) | smp_mem_inter_test | 80,984 | **1,644–1,649 ms** | **49.1–49.3 KB/s** | 1,733 µs *counted* (86.2 B) |

Per packet: *counted* = the probe's own OUT → OUT mean; the other rows divide by 9.9 bytes per packet, since OpenOCD and the images are the same.

Same image, three firmwares (smp_mem_inter_test, 80,984 bytes):

| Firmware | `load_image` time | Rate | vs. stock | vs. FreeRTOS JTAGprobe |
|---|---|---|---|---|
| Stock debugprobe 2.0.0 (computed at 1.05 KB/s) | ~77 s | 1.05 KB/s | 1× | 0.03× |
| JTAGprobe `jtagprobe-new`, FreeRTOS, 1 core | 2,380 ms | 34.0 KB/s | 32× | 1× |
| JTAGprobe_smp, first build | 8,136 ms | 9.95 KB/s | 9.5× | 0.29× |
| **JTAGprobe_smp, second build** | **2,125 ms** | **38.1 KB/s** | **36×** | **1.12×** |
| JTAGprobe_smp, third build | 2,143 ms | 37.8 KB/s | 36× | 1.11× |
| **JTAGprobe_smp, current build (512-byte packets)** | **1,644–1,649 ms** | **49.1–49.3 KB/s** | **47×** | **1.45×** |

Every one of these tests also **passed** after its load. Whole-test CTest times (15 s reset wait included), without semihosting prints (§ 9.5 has them with):

| Test | FreeRTOS JTAGprobe | JTAGprobe_smp (first build) | JTAGprobe_smp (second build) | JTAGprobe_smp (third, 10 s reset wait) | JTAGprobe_smp (current, 512-byte packets, 10 s reset wait) |
|---|---|---|---|---|---|
| smp_test4 | 29.6 s | — | — | — | — |
| smp_mem_test | 140–142 s | 61.0 s | **54.2 s** | **50.2 s** | **47.1 s** |
| smp_mem_inter_test | 190.1 s | 193.1 s | **186.0 s** | **193.4 s** | **175.9 s** |

The last two columns' runs wait 10 s after the board reset instead of 15 s. smp_mem_inter_test's time is set by its own computing on the Pi (~170 s) and varies by several seconds from run to run (186–193 s), so the probe firmware hardly shows there.

**Reading the numbers:** the first SMP build spent about **700 µs more per packet** than the single-core firmware (≈ 1,000 µs against ≈ 251–292 µs): the two cross-core wake-ups per packet of § 8.4. That made the load about **4× slower** (9.95 KB/s against 34–39.5 KB/s). The second build takes those wake-ups off the packet path: ≈ 262 µs per packet, **3.8× faster** than the first SMP build and as fast as the single-core firmware (on the same image, smp_mem_inter_test, 2.13 s against 2.38 s). The `TUD` spin of the third build changed nothing measurable (2,143 ms against 2,125 ms): § 9.4 shows why. The 512-byte packets of the current build load about **23 % faster** again (§ 9.6). smp_mem_test as a whole finished in 54.2 s with the SMP probe, against 140 s with the single-core one, so the gain there comes from the run after the load (semihosting and memory polling traffic: isolated packets rather than a stream), with the DAP work on core 1 and the USB and UART work on core 0. That part was not profiled packet by packet.

### 9.3 Computed load times per test image

Each test's loaded bytes divided by the measured rates: **39.5 KB/s** (FreeRTOS JTAGprobe, best measured), **37.9 KB/s** (JTAGprobe_smp current build, mean of its two measurements), **9.95 KB/s** (JTAGprobe_smp first build) and **1.05 KB/s** (stock). Values marked *measured* were timed; the rest are computed.

| Test image | Loaded bytes | FreeRTOS JTAGprobe | JTAGprobe_smp (current) | JTAGprobe_smp (first build) | Stock debugprobe |
|---|---|---|---|---|---|
| cmsis-os-validator | 151,640 | 3.8 s | 4.0 s | 15.2 s | 2.4 min |
| mutex-stress | 94,488 | 2.4 s | 2.5 s | 9.5 s | 1.5 min |
| rtos-apis | 228,136 | 5.8 s | 6.0 s | 22.9 s | 3.6 min |
| sd_test | 83,512 | 2.1 s | 2.2 s | 8.4 s | 1.3 min |
| smp-mat-sdcard-test | 93,736 | 2.4 s | 2.5 s | 9.4 s | 1.5 min |
| smp-mat-test | 77,608 | 2.0 s | 2.0 s | 7.8 s | 1.2 min |
| smp-num-test | 91,160 | 2.3 s | 2.4 s | 9.2 s | 1.4 min |
| smp-pipeline-test | 146,472 | 3.7 s | 3.9 s | 14.7 s | 2.3 min |
| smp-pro-cons-test | 88,968 | 2.3 s | 2.3 s | 8.9 s | 1.4 min |
| smp_mem_test | 74,888 | 1.9 s | 1.98 s *measured* | 7.54 s *measured* | 1.2 min |
| smp_mem_inter_test | 80,984 | 2.38 s *measured* | 2.13 s *measured* | 8.14 s *measured* | 1.3 min |
| smp_test0 … smp_test4 | 65,656 … 74,744 | 1.7 … 1.9 s (smp_test4: 1.78 s *measured*, 70,296 B) | 1.7 … 2.0 s | 6.6 … 7.5 s | 1.0 … 1.2 min |

**Formula:** $t_{load} \approx \frac{\text{loaded bytes}}{\text{rate}} \approx \frac{\text{loaded bytes}}{9.9} \times t_{packet}$, with $t_{packet}$ ≈ 251–292 µs (FreeRTOS JTAGprobe), ≈ 262 µs (JTAGprobe_smp second and current builds), ≈ 1,000 µs (JTAGprobe_smp first build).

### 9.4 Where the time per packet goes (third build, 64-byte packets)

From the probe's counters (§ 8.5), reset just before `load_image` and read just after it:

| Per packet, mean | smp_mem_test load (7,541 packets) | smp_mem_inter_test load (8,122 packets) | Where |
|---|---|---|---|
| OUT → next OUT (one full cycle) | **263.5 µs** | **262.8 µs** | — |
| OUT received → DAP thread starts it | 0.9 µs | 0.9 µs | core 0 → core 1 |
| `DAP_ExecuteCommand()` (JTAG bit-banging) | **68.2 µs** (max 82) | **68.3 µs** (max 81) | core 1 |
| wait for a free response slot | 0.4 µs | 0.4 µs | core 1 |
| response ready → IN started | 6.1 µs | 6.1 µs | core 1 / core 0 |
| IN started → host collected it | **72.8 µs** | **72.5 µs** | USB |
| rest of the cycle: host turnaround to the next OUT | ≈ 115 µs | ≈ 115 µs | PC: libusb + OpenOCD |
| OUT stalls (request ring full) | 0 | 0 | — |

- The hand-off between the two cores now costs about **7 µs per packet** (0.9 + 0.4 + 6.1): the dual-core split is no longer on the critical path.
- OpenOCD sends one request at a time and waits for its reply (no OUT stalls, a single packet in flight). About **70 %** of each cycle is the USB round trip and the PC's turnaround, which no split of work between the probe's cores can remove; the JTAG bit-banging is about **26 %**.
- The remaining gains are therefore fewer round trips (larger `DAP_PACKET_SIZE`, so OpenOCD packs more into each command: done in the current build, § 9.6) and faster bit-banging (PIO).

### 9.5 Semihosting load: `SEMI_PRINTS=1`

The µOS++ AArch64 port's rpi4b tests take a CMake option `SEMI_PRINTS` (0/1, hwd images only). With 1, every console line printed on the Pi's **core 0** also goes through semihosting, besides the UART; the RESULT line goes through as always. Cores 1–3 keep printing on the UART only: OpenOCD 0.12 does not resume a core other than 0 after its semihosting call on this board (the line is printed, the core stays halted, and every other core then blocks on the console lock).

| hwd test, JTAGprobe_smp | smp_mem_test `SEMI_PRINTS=0` | smp_mem_test `SEMI_PRINTS=1` | smp_mem_inter_test `SEMI_PRINTS=0` | smp_mem_inter_test `SEMI_PRINTS=1` |
|---|---|---|---|---|
| Load | 1,983 ms | 1,994 ms | 2,125 ms | 2,143 ms |
| Test run on the board | 37 s | 314 s | ~170 s | 591 s |
| CTest total | 54.2 s | **337.4 s, PASS** | 186.0 s | **620.5 s, PASS** |
| Lines through semihosting | 1 | ~300 | 1 | ~480 |

A semihosting line costs about **0.9 s**: OpenOCD's `SYS_WRITE0` reads the string from the target **one byte per JTAG memory access** (≈ 7–10 ms each), then resumes the core. That cost is OpenOCD's method, not the probe's turnaround. The table is for the 64-byte third build; with 512-byte packets (§ 9.6) both tests also pass with `SEMI_PRINTS=1`, in **308.1 s** and **561.8 s**.

### 9.6 512-byte DAP packets (current build)

Same Pi 4 B, OpenOCD and images; only `DAP_PACKET_SIZE` changed from 64 to 512 (§ 8.4, item 4). Probe counters for the load, per packet, mean:

| Per packet, mean | 64 bytes, smp_mem_test (7,541 packets) | 512 bytes, smp_mem_test (874 packets) | 512 bytes, smp_mem_inter_test (940 packets) |
|---|---|---|---|
| Loaded bytes per packet | 9.9 | **85.7** | **86.2** |
| OUT → next OUT (one full cycle) | 263.5 µs | **1,745 µs** | **1,733 µs** |
| OUT received → DAP thread starts it | 0.9 µs | 1.9 µs | 1.1 µs |
| `DAP_ExecuteCommand()` | 68.2 µs | 581.4 µs (max 602) | 582.1 µs (max 601) |
| response ready → IN started | 6.1 µs | 7.0 µs | 6.7 µs |
| IN started → host collected it | 72.8 µs | 242.9 µs | 239.7 µs |
| rest: receiving the request + PC turnaround | ≈ 115 µs | **≈ 912 µs** | ≈ 902 µs |
| Cycle time per loaded byte | 26.6 µs | **20.4 µs** | 20.1 µs |

Whole runs (CTest, 10 s reset wait), all **PASS**:

| hwd test | 64-byte packets | 512-byte packets | Change |
|---|---|---|---|
| smp_mem_test, load | 1,954–1,994 ms | 1,528–1,547 ms | −23 % |
| smp_mem_inter_test, load | 2,106–2,143 ms | 1,644–1,649 ms | −23 % |
| smp_mem_test, `SEMI_PRINTS=0` | 50.2 s | **47.1 s** | −3.1 s (−6 %) |
| smp_mem_inter_test, `SEMI_PRINTS=0` | 186.0–193.4 s | **175.9 s** | −10 s or more |
| smp_mem_test, `SEMI_PRINTS=1` | 337.4 s | **308.1 s** | −29.3 s (−8.7 %) |
| smp_mem_inter_test, `SEMI_PRINTS=1` | 620.5 s | **561.8 s** | −58.7 s (−9.5 %) |

**Reading the numbers:** - OpenOCD does fill the larger packets: **8.6× fewer packets** for the same image. - The load is still only about **23 % faster**, not 8×. The bit-banging is unchanged per byte (exec totals ≈ 510 ms for smp_mem_test in both builds), the IN transfers drop from 549 to 214 ms in total, but the *rest* of each cycle grows from ≈ 115 µs to ≈ 910 µs. That part is the reception of the request (up to eight 64-byte USB packets, each a tinyusb event on core 0 followed by a re-arm of the endpoint) plus the PC's turnaround. The counters do not separate the two, so how much of it is the per-packet re-arm is **not measured**. - The whole tests gain more than their loads (smp_mem_test −3.1 s against −0.4 s of load): the polling and semihosting traffic during the run also travels in fewer, larger packets. This part was not profiled packet by packet. smp_mem_inter_test's earlier runs spread over 186–193 s, so its gain is between 10 and 17 s. - With `SEMI_PRINTS=1` the gain is ~9 %. The bulk of those runs is still OpenOCD reading every semihosted string byte by byte (§ 9.5), which the packet size cannot change. - Next step on the probe side: time the OUT reception separately,

and arm the rest of a request in one transfer once `dap_req_len()` knows its length, instead of one 64-byte packet at a time.

## 10. Conclusion

The 2-minute download time is caused almost entirely by **synchronous debug logging ( `probe_info` ) blocking the CPU for 5.8 ms per packet over a 115,200 baud UART while the FreeRTOS scheduler is suspended**.

Applying **Patch 1** alone provides an immediate **28× speedup** (reducing 120s down to ~4s). Applying **Patches 2 and 3** brings the download time down to **under 1 second**, fully solving the performance problem while maintaining a clean, responsive hardware UART bridge on `/dev/ttyACM0`.

Measured on a Raspberry Pi 4 B over JTAG (§ 9), where OpenOCD sends about 9.9 loaded bytes per packet and one packet at a time: the optimized FreeRTOS firmware loads at **34–39.5 KB/s** against ~1.05 KB/s for the stock firmware. **JTAGprobe_smp** (µOS++ III SMP, DAP on core 1): its first build lost a factor of 4 to two cross-core wake-ups per packet (9.95 KB/s), which the second build fixed: with the DAP thread spinning on its ring and sending its own responses it loads at **37.6–38.5 KB/s**, as fast as the single-core firmware, and the complete smp_mem_test ran in 50–54 s against 140 s. The probe's own counters (§ 9.4) put the core-to-core hand-off at ~7 µs per packet and show that ~70 % of each cycle was the USB round trip and the PC. The current build therefore uses **512-byte DAP packets** (§ 9.6): 8.6× fewer packets, loads at **48.4–49.3 KB/s** (23 % faster, 1.45× the single-core firmware), and the tests finish faster (smp_mem_test 47.1 s, smp_mem_inter_test 175.9 s; with `SEMI_PRINTS=1` 308.1 s and 561.8 s, all PASS). With semihosting prints on, a semihosted line still costs ~0.9 s, set by OpenOCD (§ 9.5).

## 11. Final Reflection: One CPU vs Two CPUs

| Raspberry Pi 4 B over JTAG, OpenOCD 0.12 | 1 CPU (FreeRTOS JTAGprobe) | 2 CPUs (JTAGprobe_smp) |
|---|---|---|
| `load_image`, smp_mem_inter_test (80,984 B) | 2,380 ms (34.0 KB/s) | 2,106–2,143 ms (37.8–38.5 KB/s); **1,644–1,649 ms (49.1–49.3 KB/s)** with 512-byte packets |
| Probe time per packet (≈ 9.9 loaded bytes) | 251–292 µs | **259–264 µs** |
| Whole smp_mem_test (CTest) | 140 s | 50–54 s; **47.1 s** with 512-byte packets |
| Whole smp_mem_inter_test (CTest) | 190 s | 186–193 s; **175.9 s** with 512-byte packets |
| smp_mem_test / smp_mem_inter_test with `SEMI_PRINTS=1` | — | 337 s / 621 s; **308 s / 562 s** with 512-byte packets |

The 512-byte packets (§ 9.6) would help a single-core firmware in the same way: they cut round trips, not the hand-off between cores. The comparison of one CPU with two below is therefore made at 64 bytes, where both firmwares were measured.

### 11.1 JTAG operations

- **Bulk transfers ( `load_image` ): about the same.** OpenOCD runs a strict request/reply chain in JTAG mode: one ~64-byte command (≈ 9.9 loaded bytes), then it waits for the answer. In such a chain a second core has nothing to overlap. Of a ~263 µs cycle, the counters of § 9.4 give ~68 µs to the bit-banging on core 1, ~7 µs to the hand-off between the cores, and ~188 µs to the USB transfer and the PC's turnaround. A second core can only *lose* time here, if the hand-off sleeps: the first SMP build (history, fixed since the second build) woke the other core twice per

packet and needed ~1 ms per packet (4× slower). Keeping the packet path free of sleeps (the DAP thread spins on its ring and sends its own reply) is what made the two-core firmware as fast as the single-core one.

- **Scattered traffic (a running test): faster.** smp_mem_test, whose run is mostly OpenOCD polling the four Cortex-A72 cores between short exchanges, completed in 50–54 s instead of 140 s. The DAP work on core 1 never waits behind USB event handling or the UART on core 0. The share of each factor was not profiled. Where the test's own computing dominates (smp_mem_inter_test, ~170 s on the Pi), the probe makes no measurable difference (186–193 s against 190 s, within the run-to-run spread).

### 11.2 UART

On one CPU the UART bridge task and the DAP task share the core: the bridge has to preempt the bit-banging (it has the highest priority) or characters are lost. On two CPUs the UART thread runs on core 0, beside USB, and never delays a DAP command, nor can a DAP burst delay the UART. The tests ran with their whole console on the UART and passed; the bridge throughput itself was not benchmarked.

### 11.3 Semihosting

To the probe a semihosting call is ordinary JTAG traffic, so it gains like any scattered work. Its cost, however, is OpenOCD's: `SYS_WRITE0` reads the string one byte per JTAG memory access, ~0.9 s per console line, whatever the probe. With every core-0 line through semihosting ( `SEMI_PRINTS=1` ), smp_mem_test takes 337 s instead of 50–54 s and smp_mem_inter_test 621 s instead of 186–193 s. On this board OpenOCD also serves semihosting only from the target's core 0: a call from cores 1–3 is printed but that core is never resumed.

### 11.4 Where the next gains are

Not in another division of work between the two Pico cores (the hand-off is already ~7 µs per packet), but in:

1. **fewer round trips**: a larger `DAP_PACKET_SIZE`. **Done** in the current build (512 bytes over the 64-byte endpoint, § 9.6): loads 23 % faster, tests 6–10 % faster. What is left there is the ≈ 910 µs per packet of request reception plus PC turnaround, to be measured and, if it is the per-USB-packet re-arm, reduced by arming the rest of a request in one transfer;
2. **faster bit-banging**: a PIO state machine for TCK/TMS/TDI/TDO (now ~580 µs of each 1,740 µs cycle);
3. **the host**: reading semihosting strings in blocks instead of byte by byte.